that unfamiliar fossil species might still be living somewhere on Earth. In the late 18th century, French zoologist Georges Cuvier looked into this by exploring the anatomy of living and fossil elephants. He proved that fossil forms such as mammoths and mastodons were anatomically distinct from living elephants, so they must represent extinct species. (It was highly unlikely that they still lived on Earth without being noticed.)

Cuvier believed that Earth had experienced a series of distinct ages, each of which ended with a “revolution” that destroyed existing flora and fauna. He did not, though, believe that the evidence of fossil remains supported a theory of evolution. Nevertheless, Cuvier’s central views have continued to win support, and modern evidence points to at least five catastrophic mass extinction events in Earth’s past, including the one that wiped out the dinosaurs. Unlike Cuvier, however, today’s scientists know that life is not recreated out of nothing after a catastrophe. Rather, when a mass extinction event kills off many species, those left will evolve and multiply—sometimes relatively quickly—to fill vacant ecological niches, as the mammals did after the age of the dinosaurs.